Roll No. 263.21....

DRONACHARYA COLLEGE OF ENGINEERING, GURGAON
PRE-UNIVERSITY EXAMINATION APRIL, 2025
(GURUGRAM UNIVERSITY)

Semester: IV (B Tech/) **Branch: CSE(AIML)/ECS**

QUESTION PAPER

Subject with Code: Design & Analysis of Algorithms

Time: 3 hours **Max. Marks: 70**

Note: Q.1 is compulsory. Attempt any ONE question from each Unit. Each question carries of 14 marks. Assume data, if any missing.

Q. 1. Objective type / Short-answer type questions.

i. What are the characteristics of an algorithm? (CO1)

ii. Differentiate between Greedy and Dynamic Programming approach. (CO1)

i. Derive complexity of quick sort. (CO2)

Differentiate between Backtracking and Branch and Bound Techniques. (CO2)

Explain Greedy approach in short. (CO2)

Define disjoint sets and its two operations. (CO1)

Write the pseudo code for Divide and Conquer approach. (CO3)

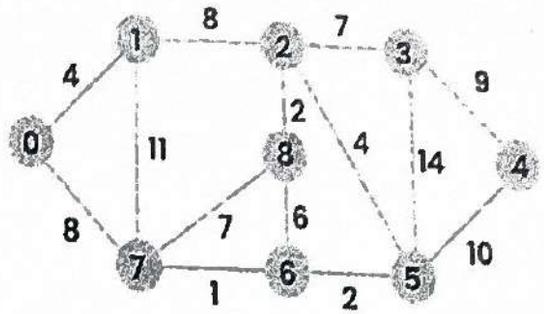
UNIT-I

a) Explain asymptotic notations? Give their mathematical models. (CO2)

Apply DAC on the algorithm of Binary Search and deduce its complexity. (CO3)

What is Knapsack problem? Write algorithm for Greedy approach to solve Knapsack. Splve the following knapsack problem using greedy approach: P(P1, P2,...,P5)= 15,12,9,16,17 and W= 2,5,3,4,6 where m=12. (CO3)

b) Solve the following using DIJKSTRA algorithm. Source is 0. (CO3)



UNIT-II

Q4. a) ***State Travelling Salesman Problem. Solve following TSP using dynamic programming*** (CO2)

1 2

$$\begin{bmatrix} 0 & 5 & 7 & 8 \\ 2 & 0 & 3 & 5 \\ 8 & 9 & 0 & 10 \\ 12 & 13 & 11 & 0 \end{bmatrix}$$

4 3

(5+5+4)

Q. 4. b) State and explain the 8-Queens problem. Write its algorithm and draw its state space tree for solution. (CO1)

Q. 5. a) Draw the portion of the state space tree generated by LCBB for the portion of the knapsack : n=5, (p1,p2,..,p5)= (10,15,6,8,4), (w1,w2,..,w5)= (4,6,3,4,2) and m=12 (CO3)

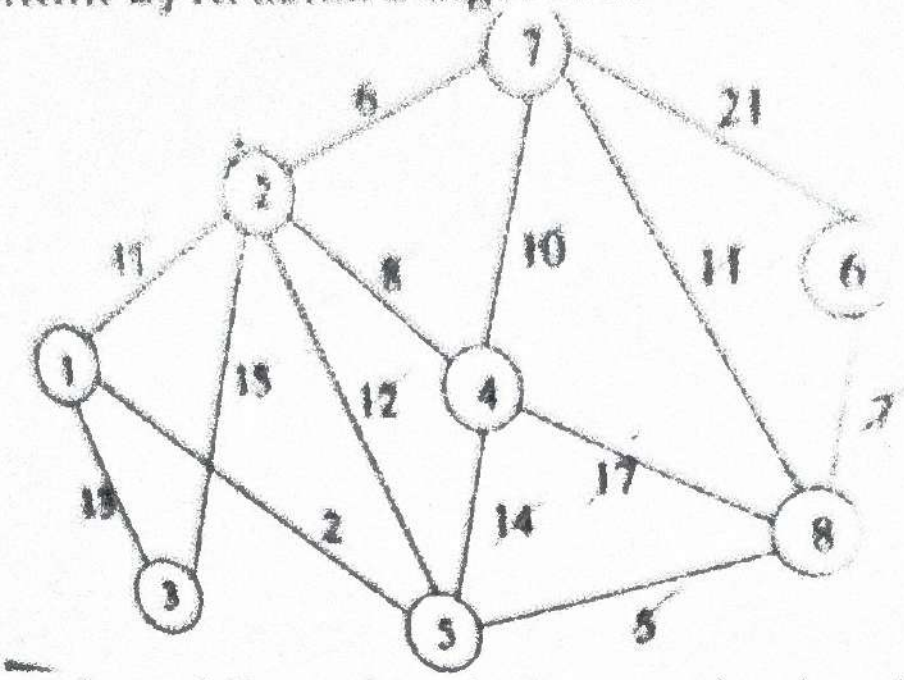
b) Deduce the complexity of merge sort with the help of an algorithm (4+5+5) (CO2

UNIT-III

Q6 a) What is graph coloring? How backtracking is used for graph coloring problem. Explain with state space tree with an example (CO3

b) Let w={5,7,10,12,15,18,20} and m=35. Find all possible subsets of w that sum to m. Draw the portion of the state space tree that is generated. (CO3)

Q7. Compute a minimum cost spanning tree for the below graph using 1) Prim's algorithm 2) Kruskal's algorithm



b) Apply Branch and Bound technique and solve the following TSP. (7+7) (CO3)

$$\begin{bmatrix} \infty & 20 & 30 & 10 & 11 \\ 15 & \infty & 16 & 4 & 2 \\ 3 & 5 & \infty & 2 & 4 \\ 19 & 6 & 18 & \infty & 3 \\ 16 & 4 & 7 & 16 & \infty \end{bmatrix}$$

UNIT-IV

Q. 8. **a)** State and prove cook's theorem. (7) (CO2)

b) Explain NP-hard graph problems. State and prove that CDP is NP-hard. (7) (CO2)

Q. 9. What is the relationship between P,NP, NP-hard and NP-complete. Show with diagrams. Also differentiate between P and NP, NP-hard and NP-complete. (CO2) (14)